

**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING**

Khwopa College Of Engineering
Libali, Bhaktapur
Department of Computer Engineering



**A PROPOSAL ON
FACIAL RECOGNITION ATTENDANCE SYSTEM
USING OPENCV**

Submitted in partial fulfillment of the requirements for the degree

BACHELOR OF COMPUTER ENGINEERING

Submitted by

Govinda Bista	KCE/077/BCT/015
Hari Krishna Khuju	KCE/077/BCT/016
Subash Bhandari	KCE/077/BCT/037
Suresh Shrestha	KCE/077/BCT/040

Under the Supervision of
Er.Dinesh Man Gothe
Department Of Computer Engineering

Khwopa College Of Engineering
Libali, Bhaktapur
2023/12/08date

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Certificate of Approval

This is to certify that this minor project work entitled “**Facial Recognition Attendance System Using OpenCV**” submitted by Govinda Bista, Hari Krishna Khaju, Subash Bhandari and Suresh Shrestha to the Department of Computer Engineering in partial fulfillment of requirement for the degree of Bachelor of Engineering in Computer Engineering. The project was carried out under special supervision and within the time frame prescribed by the syllabus.

We found the students to be hardworking, skilled, bona fide and ready to undertake any commercial and industrial work related to their field of study and hence we recommend the award of Bachelor of Computer Engineering degree.

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Er. Subhadra Joshi
(Project Supervisor)

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Er. Dinesh Man Gothe
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Abstract

There is abundance of Nepali Unicode text in internet due to social media like; Twitter and Nepali news websites. People express their feelings through text by writing reviews about products, movies, news etc. online. Markets like movie industry, tourism industry and several service providers rely on people's opinions about the products and services to get insight about the market. Manually, going through such gigantic reviews and comments is a tedious task. This document shows various approaches taken to build Nepali language sentiment classification system using recurrent neural networks which classifies the sentiment polarity of a given sentence into "positive" and "negative". The system takes input Nepali sentence and outputs the probability of sentiment classes with the best model achieving the accuracy of 82%.

Keywords: *Natural Language Processing, Nepali Language, Neural Network, Machine Learning, Sentiment Classification*

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List of Symbols and Abbreviation

NLP	Natural Language Processing
RNN	Recurrent Neural Network
ARNN	Attention Based Recurrent Neural Network
JS	JavaScript
RegEx	Regular Expressions
JSON	JavaScript Object Notation
KB	Kilo Byte
OOV	Out of Vocab
NELRALEC	Nepali Language Resources and Localization for Education and Communication.
NNC	Nepali National Corpus
POS	Parts of Speech
LSTM	Long Short-Term Memory